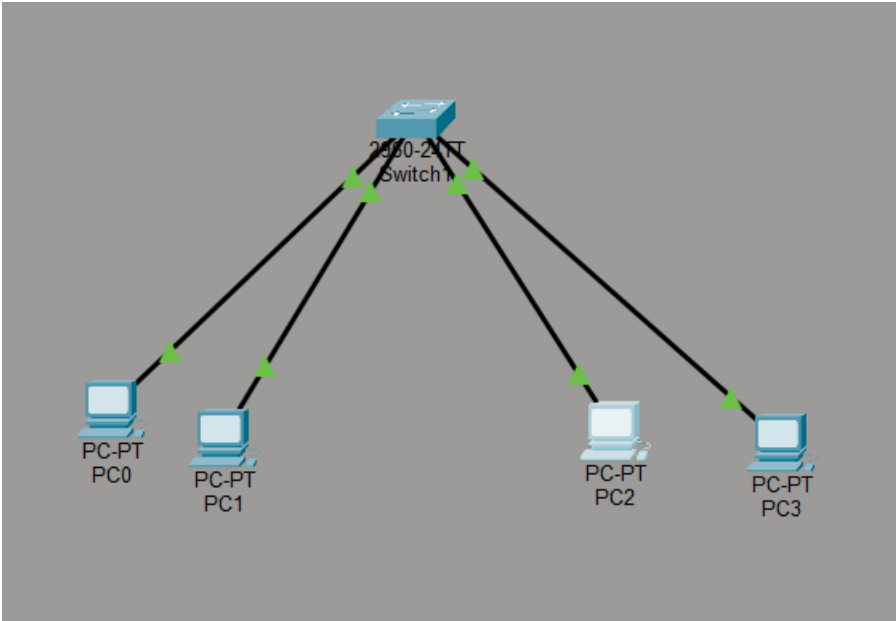


**Q-Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: Use ping or file sharing to test connectivity between devices in different VLANs.**

To create two VLANs on a switch, assign devices to each VLAN, and verify that devices within the same VLAN can communicate while communication between different VLANs is blocked.

Network Topology:



PC IP Address

Device	VLAN	IP Address
PC0	VLAN 10	192.168.10.2
PC1	VLAN 10	192.168.10.3
PC2	VLAN 20	192.168.20.2
PC3	VLAN 20	192.168.20.3

Switch Configuration

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Workstations
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Smart_devices
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Workstations	active	Fa0/1, Fa0/2
20	Smart_devices	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0

--More--

Connectivity Test

Test 1: Same VLAN Communication

From PC0:

ping 192.168.10.3

Result: **Successful**

From PC2:

ping 192.168.20.3

Result: **Successful**

Test 2: Different VLAN Communication

From **PC0**:

ping 192.168.20.2

Result: **Request timed out**

From **PC1**:

ping 192.168.20.3

Result: **Request timed out**

Two VLANs, **VLAN 10 (Workstations)** and **VLAN 20 (Smart Devices)**, were successfully created on the switch. Two devices were assigned to each VLAN. Devices within the same VLAN communicated successfully using the ping command, while communication between different VLANs was blocked because no inter-VLAN routing was configured.